

## SANDHEADS, BC, Canada

WMO: 712090

Lat: 49.1058N

Lon: 123.3033W

Elev: 36

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 26.7       | 30.2 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 36.6                  | 46.4 | 33.6 | 45.8 | 13.6                  | 110  | 0.637 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 7.6                    | 72.3            | N/A  | 70.3 | N/A  | 68.7 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 9.0                  | 300  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 30.4          | 26.8 | 23.9 | DB                         | 22.8 | 77.2               | 3.9 | 2.6                                                | 20.0 | 79.1       | 17.7 | 80.6       | 15.6 | 82.0       | 12.8 | 83.9 | (4) |
| (5)               |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A  | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 51.7   | 41.6  | 42.3  | 45.2  | 49.4  | 55.2  | 59.5  | 64.1  | 64.2  | 59.5  | 52.0  | 45.5  | 41.1 |
|                                                                               | DBStd       | 9.24   | 5.10  | 4.30  | 3.58  | 3.40  | 3.74  | 3.55  | 3.24  | 2.84  | 3.35  | 3.40  | 4.97  | 5.06 |
|                                                                               | HDD50       | 1125   | 262   | 217   | 151   | 51    | 3     | 0     | 0     | 0     | 0     | 19    | 146   | 276  |
|                                                                               | HDD65       | 4917   | 727   | 637   | 612   | 469   | 303   | 170   | 56    | 49    | 166   | 402   | 586   | 740  |
|                                                                               | CDD50       | 1746   | 1     | 1     | 4     | 31    | 166   | 286   | 436   | 442   | 286   | 81    | 11    | 1    |
|                                                                               | CDD65       | 62     | 0     | 0     | 0     | 0     | 0     | 6     | 28    | 25    | 3     | 0     | 0     | 0    |
|                                                                               | CDH74       | 22     | 0     | 0     | 0     | 0     | 0     | 2     | 15    | 4     | 1     | 0     | 0     | 0    |
|                                                                               | CDH80       | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 11.4   | 11.9  | 11.8  | 12.3  | 11.4  | 10.5  | 10.5  | 10.9  | 10.1  | 10.7  | 11.3  | 12.8  | 12.9 |
| Precipitation                                                                 | PrecAvg     | 70.2   | 9.9   | 6.4   | 6.7   | 4.7   | 3.4   | 2.8   | 1.8   | 2.3   | 3.4   | 7.9   | 10.7  | 10.2 |
|                                                                               | PrecMax     | 87.4   | 16.8  | 12.8  | 12.6  | 9.7   | 5.4   | 5.1   | 3.9   | 10.0  | 7.5   | 14.3  | 19.4  | 15.1 |
|                                                                               | PrecMin     | 55.4   | 5.3   | 0.7   | 1.8   | 1.5   | 0.6   | 0.6   | 0.3   | 0.4   | 0.6   | 1.4   | 4.9   | 5.9  |
|                                                                               | PrecStd     | 7.9    | 2.7   | 2.5   | 2.2   | 1.9   | 1.2   | 1.3   | 0.9   | 2.0   | 1.9   | 3.1   | 3.7   | 2.2  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 52.9  | 52.5  | 56.1  | 60.9  | 67.6  | 72.5  | 75.8  | 74.3  | 70.8  | 61.5  | 57.1  | 53.0 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 2%          | DB     | 50.7  | 50.3  | 52.8  | 57.5  | 64.5  | 69.4  | 72.7  | 72.0  | 67.3  | 59.3  | 54.5  | 50.5 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5%          | DB     | 49.3  | 48.6  | 51.1  | 55.4  | 62.3  | 66.7  | 70.6  | 70.1  | 65.3  | 57.6  | 53.0  | 48.6 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 10%         | DB     | 47.9  | 47.3  | 49.7  | 53.8  | 60.3  | 64.5  | 68.8  | 68.5  | 63.8  | 56.3  | 51.5  | 47.2 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 5.4   | 5.5   | 6.1   | 6.7   | 7.7   | 7.8   | 8.5   | 7.6   | 5.9   | 4.9   | 5.3   | 5.4  |
|                                                                               |             | MCDBR  | 6.4   | 6.5   | 7.3   | 8.2   | 10.1  | 10.5  | 10.4  | 9.5   | 7.8   | 6.2   | 6.6   | 6.7  |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.320  | 0.313 | 0.331 | 0.361 | 0.382 | 0.360 | 0.355 | 0.359 | 0.343 | 0.335 | 0.321 | 0.309 |      |
|                                                                               | taud        | 2.476  | 2.536 | 2.502 | 2.418 | 2.382 | 2.462 | 2.505 | 2.503 | 2.550 | 2.552 | 2.518 | 2.483 |      |
|                                                                               | Ebn at Noon | 235    | 269   | 281   | 281   | 279   | 285   | 285   | 279   | 274   | 257   | 233   | 223   |      |
|                                                                               | Edh at Noon | 20     | 23    | 28    | 34    | 36    | 34    | 32    | 31    | 27    | 23    | 19    | 17    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 275    | 548   | 828   | 1284  | 1675  | 1802  | 2008  | 1702  | 1183  | 650   | 325   | 220   |      |
|                                                                               | RadStd      | 44     | 82    | 112   | 136   | 143   | 208   | 172   | 130   | 105   | 87    | 59    | 33    |      |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) Regional (68 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page